

VotaSJ: Trust, Counted Live

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Abstract. *VotaSJ is a blockchain-anchored participatory-budgeting platform for São José/SC. Citizens authenticate with gov.br, vote from their phone, and audit the tally and the budget execution in real time — without having to trust the City Hall to count its own vote. The platform replaces a BRL 300–500k physical process with a BRL 90–160k digital one, governed by a 3-of-5 multisig (City Hall, UFSC, TCE-SC) and a 14-day timelock on rule changes. This short paper makes the case to the institutional buyer (§??) and to the citizen (§??).*

For the Investor and the Municipality

Problem. Brazilian municipalities run participatory budgets that almost no one trusts. In São José/SC: turnout below 2 % of 180,000 voters (<3,600 people decide tens of millions of reais); operational cost of **BRL 300–500k/cycle** dominated by physical assemblies and ≈3-week manual tallying; and a tally run, counted, and announced by the City Hall itself, so every contested result ends in an unfalsifiable accusation of manipulation.

Solution. A blockchain-anchored platform on Polygon PoS. Citizens authenticate with gov.br, vote from their phone, and audit results live. *On-chain:* eligibility commitments, votes, tally, budget-execution milestones. *Off-chain:* personal data, proposal media (IPFS), and UX. The blockchain is the substrate of record only where operator honesty cannot be assumed. Administration is a **3-of-5 multisig** (3 City Hall, 1 UFSC, 1 TCE-SC). Contract upgrades are gated by a **14-day timelock**. The City Hall cannot unilaterally change rules or rewrite results; the chain keeps running even if it stops cooperating.

Why now. gov.br is mature (≈150 M citizens, free for municipal integration); Polygon per-vote gas is under USD 0.005; OpenZeppelin + Hardhat make audited deployments affordable; TCE-SC and MP press for auditable processes — we arrive with the audit built in.

Market.

Segment	Scope	3y addressable
São José/SC (anchor)	≈180k voters	BRL 220k/yr
Greater Florianópolis	4 cities, ≈1.1M voters	BRL 0.8–1.2M/yr
Santa Catarina state	≈5.5M voters	BRL 4–6M/yr
BR cities >50k inhab.	≈340 cities, ≈85M voters	BRL 60M+/yr

TAM: >340 cities legally required to run participatory cycles. SAM: ≈80 with budgeted digital modernization. SOM (36 mo): single-digit municipalities anchored on São José.

B2G SaaS model. Annual subscription BRL 180–250k/city; per-cycle setup BRL 40k; implementation consulting BRL 60–120k; white-label for professional councils. Citizens

never pay. *Unit economics*: physical process **BRL 300–500k/cycle**; VotaSJ **BRL 90–160k/cycle, all-in**; net savings **BRL 170–300k/cycle**, before counting the political value of an unimpeachable result.

Traction. Production-ready platform — VoterRegistry, ParticipatoryBudget, and ExecutionTracker deployed under a UUPS proxy with a 3-of-5 multisig and 14-day timelock, exercised by a 40-test Hardhat suite, CI-enforced lint/coverage gates, $\approx 430k$ gas measured for a full citizen flow. Institutional partners: UFSC Blockchain Lab, São José Planning Secretariat, TCE-SC. Rollout: field pilot 2026-Q3, full São José cycle 2026-Q4, multi-city expansion 2027.

Ask. An institutional pilot agreement with São José/SC and a seed envelope of **BRL 400–600k** for the first external smart-contract audit, gov.br integration certification, and the field-pilot cycle. First paid contract closes 12 months after disbursement; payback is one cycle.

For the Citizen

Your vote, your eyes. Today, when you participate in São José’s participatory budget — if you participate — you go to a night-time assembly, raise your hand, and trust that the City Hall counts correctly and executes what won. VotaSJ changes one thing: **you can stop trusting and start checking.**

What changes for you.

- **Vote from your phone, on your own schedule.** No more assemblies you can’t attend because of work, kids, or distance.
- **See your vote land.** The moment you cast it, the public tally updates. Refresh the page and watch it.
- **Anyone audits the count.** Journalists, neighborhood associations, TCE-SC, MP — same data, same chain, same time.
- **Follow execution.** You’re notified when your winning proposal moves to contract, work start, milestones, and finish, with evidence you can check.
- **Your data stays private.** CPF, name, address: none of it goes on the blockchain. Only a one-way hash that proves you’re eligible, not who you are.

Three steps. (1) Sign in with gov.br — the federal ID you already use. (2) Pick a proposal: browse what neighbors submitted, read the budget, vote once. (3) Watch the result and the execution: real-time tally and per-milestone updates. No need to know what a “blockchain” is. No crypto, no fee, no extra app.

Forquilha, next cycle. Maria, a cashier whose shift ends at 22:30, opens VotaSJ on the bus home and votes for the renovated daycare. Months later she is notified — contract signed, work started, daycare reopens — each step with verifiable photographic evidence. 90 seconds of participation; the institution actually delivered.

Trust, but verify — and now you actually can.

Companion document and references. A 16-page technical white paper covers architecture, threat model, gas measurements, team, and roadmap. Sources: *Estatuto da Cidade* (https://www.planalto.gov.br/ccivil_03/leis/leis_2001/110257.htm), *LGPD* (https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/lei/113709.htm), *gov.br* (<https://acesso.gov.br/roteiro-tecnico/>), *Polygon PoS* (<https://docs.polygon.technology/>).